

FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

Amirreza Mottafegh[†], Mincheol Park[†], [Name to be added], Dr. Myeong, Professor Park, and Dr. Gwang-Noh Ahn

Digital Chemical Research Center, Korea Research Institute of Chemical Technology, 141Gajeongro, Yuseong, Daejeon 34114, Republic of Korea

[†]Amirreza Mottafegh and Mincheol Park contributed equally to this work. Author names, affiliations, and the additional author will be finalized before submission.

ABSTRACT

Translating a batch protocol into continuous flow requires chemical reasoning, engineering calculations, and equipment choices to remain consistent. We present FlowPilot, a multi-agent AI system that connects these decisions through standardized chemist intake, literature retrieval, deterministic calculation, specialist review, and inventory reconciliation. A corpus of 464 classified literature records and 2,537 handbook-derived rules supports the workflow. The resulting design record links stage conditions, stream assignments, equipment, and the process diagram. We evaluate the architecture using matched one-shot and full-pipeline generation across three chemistry cases, five retained generator models, and three repeats, yielding 90 outcomes. Separate LLM judges apply 14 fixed criteria informed by deterministic checks. The mean benchmark score increases from 0.794 for one-shot generation to 0.917 for FlowPilot, while raw judge critical flags decrease from 98 to 4. Internal ablations examine the contribution of council review. Photoredox Giese addition followed by aerobic oxidation and telescoped DPDTC-mediated amidation are selected for prospective laboratory evaluation. The results support an auditable route from a batch protocol to a first flow experiment, while leaving reaction performance to experimental validation.

INTRODUCTION

Continuous flow chemistry has matured into one of the main technologies¹⁻² for pharmaceutical³, fine-chemical⁴, and on-demand manufacturing⁵, offering orders-of-magnitude improvements in heat and mass transfer⁶, intrinsically safer handling of hazardous intermediates⁷, access to short residence times², and

ready accommodation of photochemistry⁸, electrochemistry⁹, liquid-phase oxidation¹⁰, and multi-step coupling¹¹⁻¹³. These features can reduce solvent use, energy demand, and equipment footprint when the process is appropriately designed, supporting the resource-efficiency goals of sustainable manufacturing. A mature commercial ecosystem now supplies virtually everything a flow chemist needs¹⁴, yet adoption remains gated less by what flow can do than by what it costs to design: the gap between a published batch protocol and a working flow process is typically closed by a small team over weeks of iteration; even within established translation paradigms such as microwave-to-flow, each new chemistry class typically requires re-derivation of operating conditions¹⁵. The bottleneck has shifted from running flow chemistry to designing flow processes. Multistep apixaban synthesis¹⁶ and integrated photochemical synthesis and separation of sartan intermediates¹⁷ illustrate the practical importance of coordinating reaction development with downstream operations.

Designing a flow process is a chain of interlocking decisions across fluid dynamics¹⁸⁻¹⁹, mass transfer²⁰, heat transfer²¹, materials compatibility, kinetics, and safety, where each decision constrains the next²²⁻²³. A change in feed concentration affects stoichiometry and throughput; an interstage addition changes the flow entering the next reactor; and a reactor choice constrains achievable volume, pressure, and irradiation. Existing tools address different parts of this problem. Process simulators²⁴ support engineering analysis, including extensions toward automated flowsheet design²⁵; Bayesian and active-learning approaches²⁶⁻³⁰ optimize experimental conditions; and retrosynthesis tools³¹⁻³³ address route planning. Handbook knowledge^{22-23,34} supplies practical flow-design principles, but is distributed across sources and often expressed qualitatively. The gap addressed here is their integration: translating a chemistry brief into a coherent first flow proposal that can be checked against the laboratory's actual equipment.

Recent large-language-model (LLM) copilots for chemistry³⁵⁻³⁶ have expanded what AI can do for a chemist, from synthesis planning to autonomous experimentation³⁷⁻⁴⁰. Flow-specific developments include UV-visible auto-reactometry coupled to LLM-assisted Bayesian optimization for photochemical screening⁴¹. Such systems address condition optimization, whereas the task considered here is to construct and reconcile an equipment-constrained process proposal from a batch procedure. When a monolithic LLM is prompted directly for a flow reactor design, it must handle chemistry, kinetics, fluid mechanics, heat transfer, and safety within a single inference pass⁴²⁻⁴³, without a separate calculation and reconciliation workflow. A fluent answer therefore does not by itself establish cross-domain consistency or show which constraints were checked.

The structure of flow process design suggests a division of responsibility: LLMs interpret chemistry and propose operating choices, deterministic code calculates dependent quantities, and specialist agents

review assumptions across domains. The organizing principle is therefore not simply to use more agents, but to make chemical reasoning and engineering decisions refer to the same process. This leads to the central question of the study: under the same chemistry and inventory constraints, does this coordinated workflow produce more consistent and complete flow designs than one-shot generation by the same model?

We introduce FlowPilot, a multi-agent AI pipeline purpose-built for continuous flow chemistry process design. Given a batch protocol, the system returns a flow-process proposal accompanied by a process diagram, traceable engineering calculations, literature analogies, and explicit uncertainty flags. The workflow couples five operations: (i) standardized intake that records the protocol, objective, historical evidence, inventory, and chemist hypotheses; (ii) chemistry interpretation and plan-aware retrieval from a curated flow-chemistry corpus; (iii) deterministic engineering calculations supported by handbook-derived rules; (iv) a seven-agent council that reviews and revises candidate designs under bounded domain authority; and (v) inventory reconciliation and rendering from a common final design record. Candidate selection considers the chemist's objective and process constraints. Intensification is evaluated where justified, rather than treated as evidence that any shorter residence time will preserve reaction performance.

The evaluation follows that question from information quality to design quality. We first characterize the literature corpus and engineering rule base, then test whether chemistry-aware retrieval improves the relevance of selected analogies. We next compare the delivered one-shot and FlowPilot designs using a common fixed rubric across local and commercial models. Module ablations examine the role of council review, while an earlier isoxazole cycloaddition model-matrix study⁴⁴ characterizes configuration sensitivity. Finally, photoredox Giese addition followed by aerobic oxidation⁴⁵ and telescoped DPDTC-mediated amidation⁴⁶ provide complementary cases for prospective evaluation of connected laboratory processes. Retrieval relevance, computational design consistency, and measured chemical performance are thus treated as distinct levels of evidence.

RESULTS

FlowPilot organizes batch-to-flow translation around a shared process description (Figure 1). The chemist supplies the protocol, design objective, and laboratory inventory, then resolves standardized follow-up questions about missing chemistry, constraints, and experimental history. The confirmed answers form a stored input package for the run. The upstream module converts that context into a chemistry plan identifying species, reaction stages, stream incompatibilities, sensitivities, and possible flow-specific

advantages. Retrieval supplies relevant precedents, and engineering calculations establish an initial proposal containing reactor geometry, stream assignments, residence-time assumptions, and operating conditions. Council review and equipment reconciliation then assess that proposal before it becomes the final reported design.

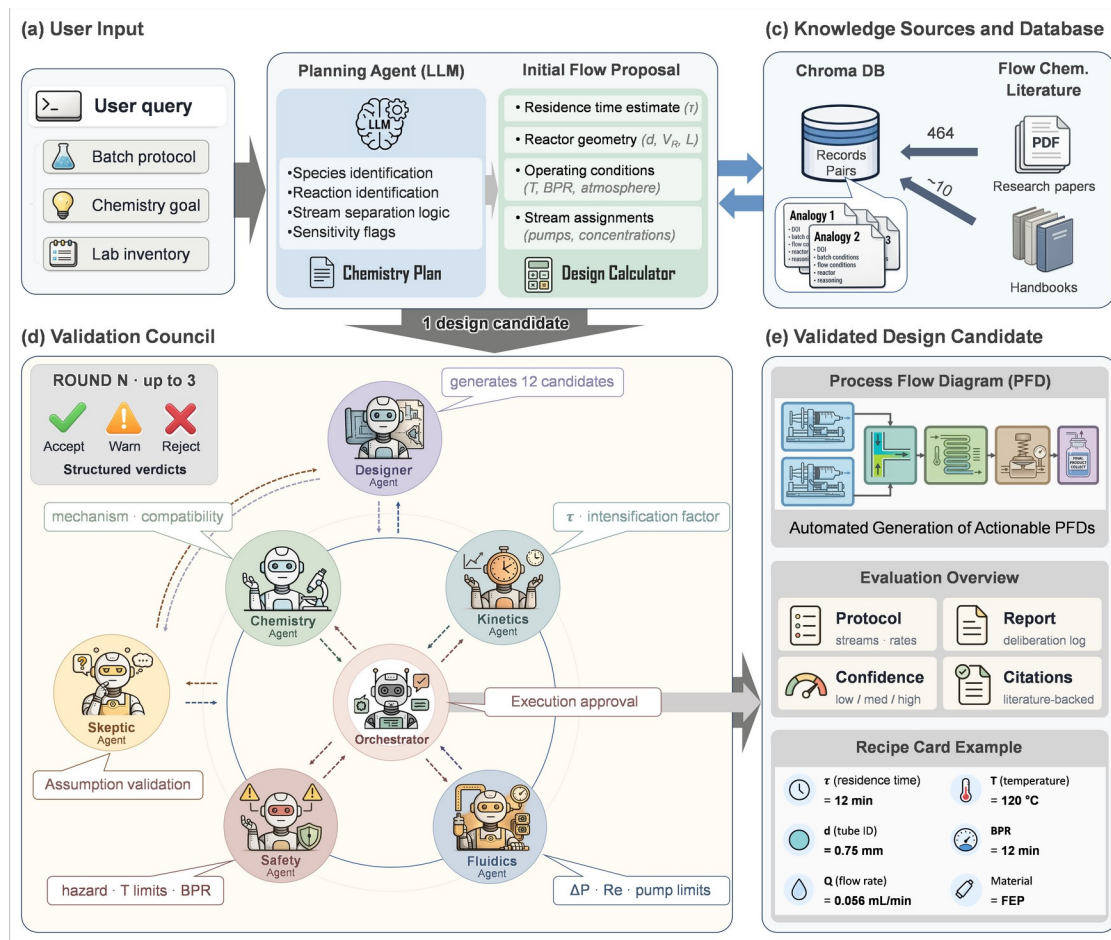


Figure 1. FlowPilot system architecture. User inputs are converted into a chemistry plan and an initial engineering proposal, supported by flow-chemistry literature and handbook knowledge. The seven-agent council generates and reviews candidate designs, records structured verdicts, and proposes bounded revisions. The selected candidate is reconciled with engineering calculations and laboratory inventory before a process-flow diagram, protocol, report, confidence assessment, and supporting citations are returned. The illustrated recipe card is schematic; computational validation does not establish experimental performance.

The downstream council supplies chemical and cross-domain review alongside deterministic numerical checks (Figure 1d). Rather than asking one model to produce a single final answer, the system generates a

candidate pool and distributes review across domain-specialized agents. The Designer proposes candidate geometries and operating points; Chemistry Agent evaluates chemical compatibility, photochemical constraints, and stream logic; Kinetics Agent evaluates residence time and conversion plausibility; Fluidics Agent evaluates flow regime, pressure drop, mixing, and hardware compatibility; Safety Agent evaluates pressure, thermal, material, and handling risks; the Skeptic audits assumptions and internal consistency; and the Orchestrator selects the final candidate. This structure produces not only a selected design but also a record of rejected alternatives, revised candidates, disqualified candidates, and the rationale for each decision.

For a reconciled result, the output is a structured design package containing stage and stream parameters, equipment assignments, a process diagram, supporting analogies, and the review record. Numerical views and the diagram refer to the same final design contract, while unresolved requirements remain visible. This common record connects a chemical proposal to specific equipment and makes subsequent experimental feedback attributable to the conditions **actually considered**. The **authority boundaries, input contracts, and provenance** mechanisms **are** described in Figures S1-S2 and Tables S1-S3; computational closure does not itself establish reaction performance.

Literature and rule-base coverage

The literature dataset covers a broad range of flow-chemistry applications (Figure 2). The curated export contains 464 classified literature records spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types range from capillary/coil and microreactor/chip systems to photoreactors and packed beds, with fluoropolymer tubing prominent among the reported materials. Explicit inlet-stream counts are available for 154 processes; two- and three-stream configurations account for 91 and 30 of these records, respectively. These annotations supply precedents rather than prescribing a default number of streams. The diversity supports retrieval across photochemical and non-photochemical processes with different materials and operating regimes.



Figure 2. Literature corpus and engineering rule-base coverage. (a-d) Reaction classes, reactor types, reported materials, and inlet-stream configurations in the curated flow-chemistry corpus; explicit inlet-stream counts are available for 154 records. (e) Category and severity distribution within the 2,537-rule engineering knowledge base. (f) Recorded associations between rule categories and chemistry classes. Full categorical exports and field-specific availability are reported in Figures S3-S4 and Tables S4-S5; associations indicate knowledge coverage rather than independent validation of every rule.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (Figure 2e-f). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape supplies domain context for

evaluating model-generated designs alongside deterministic checks, rather than relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in Figures S3-S4 and Tables S4-S5.

Plan-aware retrieval improves analogy quality

Coverage alone does not show that a relevant precedent will be selected. We therefore evaluated the retrieval system against a semantic-search baseline (Figure 3). The baseline embeds the batch protocol or query summary and ranks records by cosine similarity. FlowPilot enriches the query using the chemistry plan, including reaction class, mechanism, photocatalyst family, solvent, and wavelength. Filtering starts with paired records under mechanism and phase constraints, relaxes filters when fewer than three matches remain, and expands to all records if no paired match remains. A field-aware score then reranks the candidates. Figure 3a summarizes the workflow; Figure S5 and Table S6 specify its tier transitions and scoring weights.

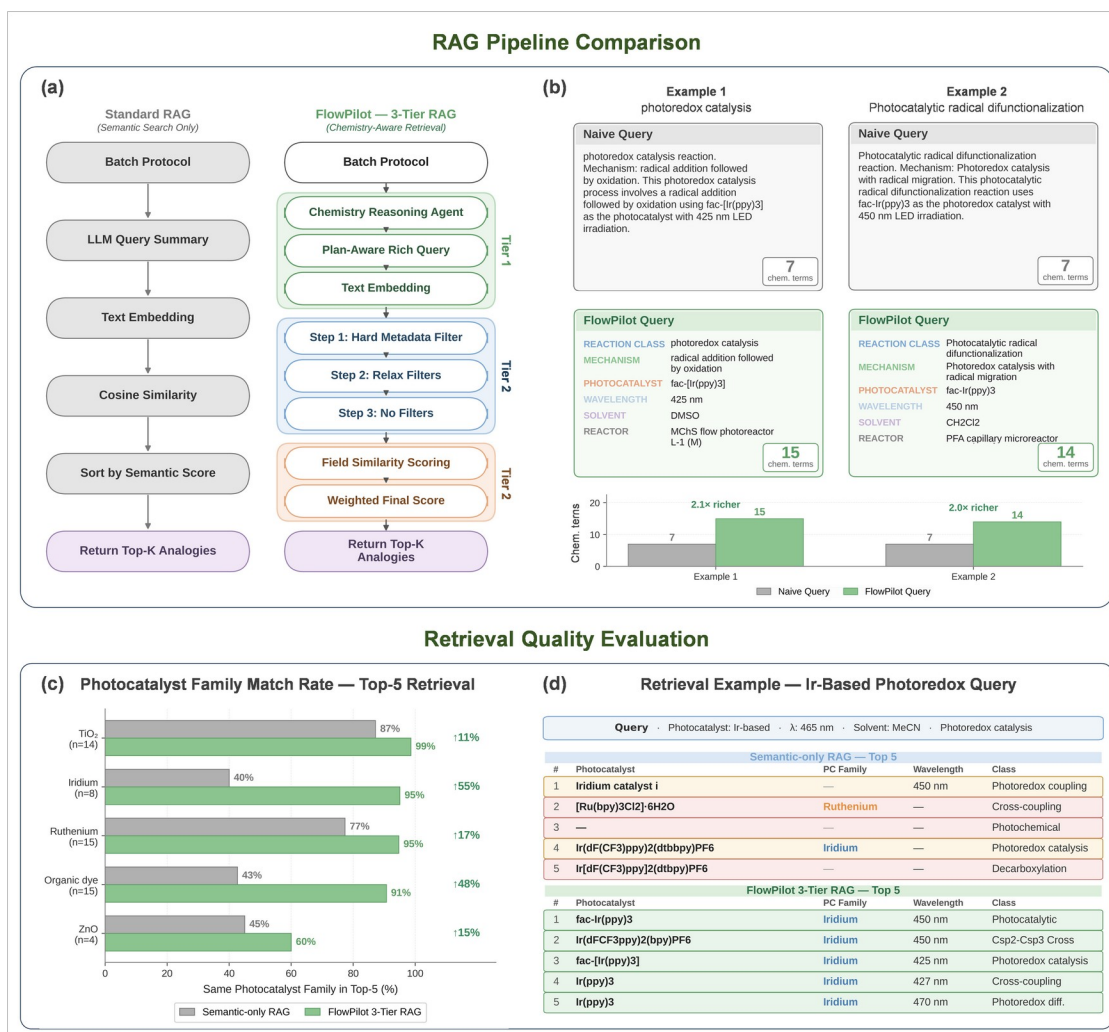


Figure 3. Plan-aware retrieval. (a) Three-tier filtering strategy: hard metadata filter, relaxed metadata filter, and pure semantic fallback. (b) Query enrichment expands chemically specific search terms approximately two-fold relative to a naive semantic baseline. (c) Top-5 photocatalyst-family matching rates by family (iridium, organic dye, ruthenium, TiO₂, ZnO) for semantic baseline vs FlowPilot reranking. (d) Representative reranking outcome showing closer wavelength and photocatalyst-family alignment.

The enriched query representation approximately doubled the number of chemically specific search terms in the two examples shown in Figure 3b. A photoredox-catalysis query increased from 7 naive terms to 15 FlowPilot terms, and a photocatalytic radical difunctionalization query increased from 7 to 14 terms. In the retrieval-pair benchmark, 1,600 query–result pairs were evaluated. FlowPilot reranked 25.1% of these

1,600 pairs relative to the pure semantic baseline, indicating that field-aware scoring changed the ordering of the candidate analogies presented to the design engine.

The strongest retrieval improvement was observed for photocatalyst-family matching. For iridium systems, top-five family matching increased from 40.0% under semantic search to 95.0% after reranking. Organic-dye systems improved from 42.7% to 90.7%, ruthenium systems from 77.3% to 94.7%, TiO₂ systems from 87.1% to 98.6%, and ZnO systems from 45.0% to 60.0% (Figure 3c). The demonstration in Figure 3d illustrates how field-aware ranking promotes records with a matching photocatalyst family and relevant wavelength context. These results measure metadata alignment, not the accuracy of a transferred kinetic estimate or the eventual reaction yield. Retrieval tiers, scoring weights, benchmark data, and source-exclusion controls are detailed in Figures S5-S6 and Tables S6-S7.

Ablation and architecture comparison

Improved retrieval does not by itself establish that the resulting flow design is consistent. We therefore compared one-shot generation with the full FlowPilot pipeline while holding the chemistry, inventory, and requested output contract constant (Figure 4a). The three cases were CuAAC, photochemical oxidation, and hydrogenolysis, with three generation repeats per model-architecture-case combination. Separate Qwen, OpenAI, and Claude judge calls evaluated normalized final designs against 14 fixed criteria, using the same source context and deterministic verification evidence. Criteria address chemical fidelity, stream and reactor consistency, equipment feasibility, operating procedure, and uncertainty. Applicable integer ratings from 0 to 4 were averaged with equal criterion and judge weights and normalized to a score from 0 to 1. Critical flags were recorded separately. Tables S8-S10 provide the scoring anchors, criterion definitions, and module-screen summaries.

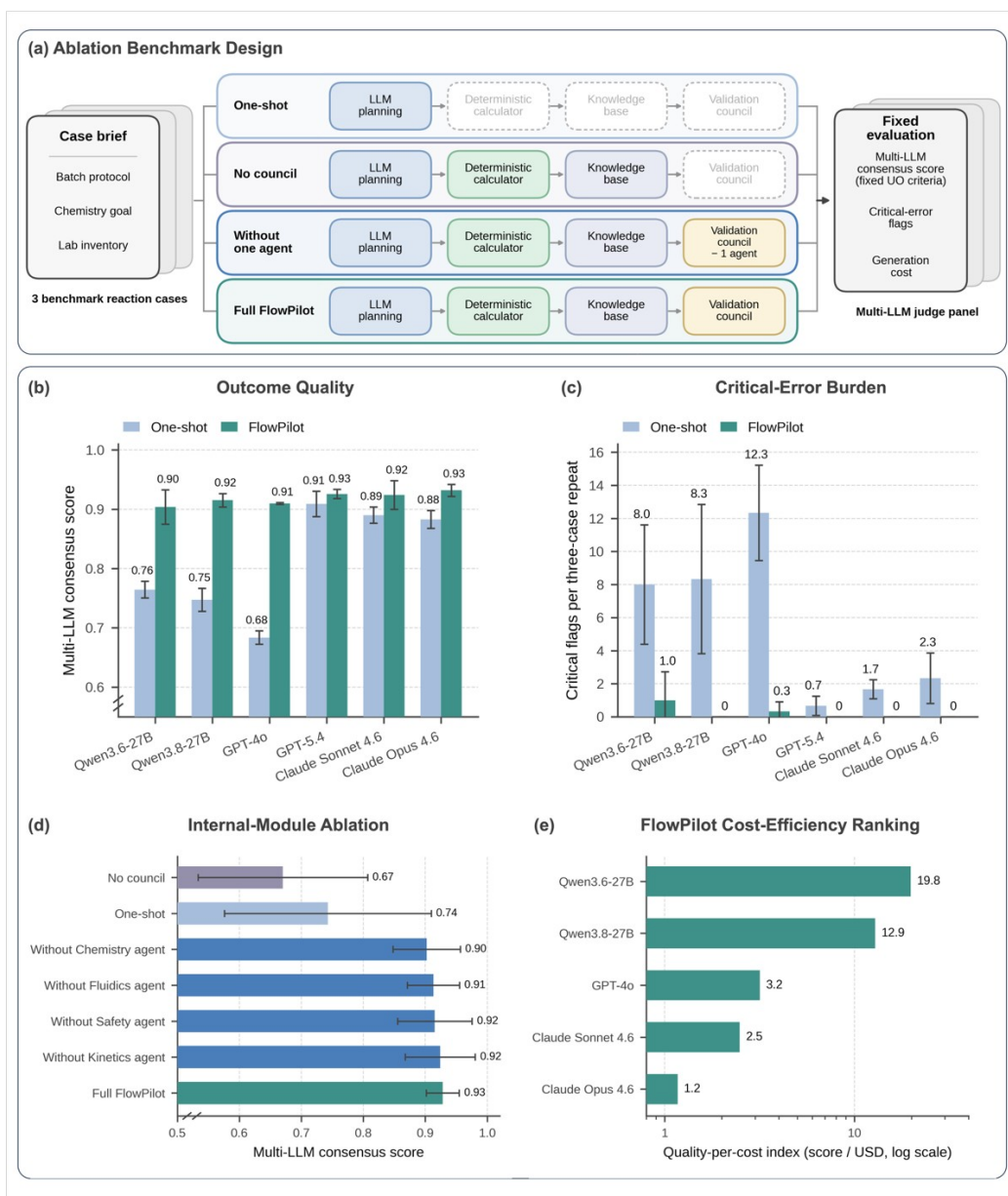


Figure 4. Ablation benchmarking and cost efficiency. (a) Architecture comparison under shared inputs and fixed evaluation. (b) Benchmark scores and (c) raw judge critical flags: mean and sample SD across three three-case repeats. Flags are not independently confirmed physical errors. These panels retain the archived GPT-5.4 comparison, whereas the ESI summary cohort contains five models. (d) Selected Qwen3.8-27B module ablations: mean and between-case SD, with one generation per case-condition. (e) FlowPilot score per generation cost under the recorded price schedule (log scale), excluding deployment costs. Full definitions and case-level results are in ESI Section 3.

Across the five-model cohort in Table S11, full-pipeline mean scores ranged from 0.904 to 0.932, compared with 0.684 to 0.890 for matched one-shot generation (Figure 4b). The mean gain was largest for GPT-4o (+0.226), followed by Qwen3.8-27B (+0.168) and Qwen3.6-27B (+0.139); the Claude gains were smaller (+0.034 and +0.049). The judges recorded 98 critical flags across the 45 one-shot outcomes and 4 across the 45 FlowPilot outcomes; 18 and 43 outcomes, respectively, had no critical flag (Figure 4c; Table S12). The score and flag summaries therefore point in the same direction within this cohort: improved completeness and fewer flagged inconsistencies. They do not establish universal model superiority or measured reaction success. Figure 4d provides a complementary internal comparison, with council removal causing the largest score decrease among the configurations shown.

The aggregate comparison is complemented by chemistry-specific scores (Figure S7), criterion-level score differences (Figure S8), and the location of critical flags within individual runs (Figure S9). Figure S10 reports tokens, estimated generation costs, and observed runtime. The higher Qwen score-per-generation-cost values in Figure 4e therefore reflect the recorded token-price schedule, not a measurement of total deployment cost. Tables S13-S14 define numerical closure and equipment checks; Tables S15-S16 specify feedback and replay records. Figure S11 and Table S17 distinguish user-composed input from system-assembled prompts and returned artifacts for one matched run. This supports the practical input-handling argument without implying that FlowPilot uses fewer total model tokens. The legacy heuristic QA values in that worked example are not the fixed-criteria scores used in Figure 4.

The earlier council model-matrix and candidate-budget experiment is reported separately in Figure S12 and Tables S18-S21. In the five-repeat comparison, the normalized radar-area score changed from 0.31 +/- 0.03 before council review to 0.48 +/- 0.23 afterward. This engineering-profile measure is not the multi-LLM benchmark score in Figure 4. The ESI defines Need revision, New versions, Combined pool, and Disqualified, and reports runtime, token use, geometry, and radar normalization. Figure S13 and Table S22 distinguish the final structured FlowPilot topology from its contemporaneous generated narrative; the complete one-shot response is retained alongside this worked example.

The ESI also shows how the chemist supplies and inspects the information behind a design. Figures S14-S15 and Table S23 document protocol entry and fixed-ID follow-up questions. Figure S16 and Table S24 show inventory import, normalization, and equipment warnings; Figure S17 and Table S25 show stage-resolved topology and stream parameters from a reopened run. Concrete failures are retained in Table S26, and Figure S18 and Table S27 expose actual council disagreement and selection records. Figure S19 and Table S28 place these views in the complete GUI and saved-run context. These operational records

connect the computational benchmark to an inspectable workflow. The remaining question, addressed by the planned cases below, is how selected proposals perform in connected laboratory reactions.

Case study 1: photoredox Giese addition followed by aerobic oxidation

The first connected-process case couples photoredox Giese addition with oxidation of the resulting sulfide to a sulfoxide (Figure 5). In the literature transformation, photoinduced single-electron transfer generates an alpha-thiomethyl radical from an alpha-silyl sulfide; addition to acrylonitrile constructs the C-C bond, and subsequent oxygen-dependent photochemistry changes the sulfur oxidation state.⁴⁵ This sequence tests more than residence-time selection: the oxygen-free first stage and oxidizing second stage must remain distinct within a continuously connected process. The collaborator's batch protocol uses 4 h under argon followed by 6 h exposed to air. These durations define source observations, not a kinetic law that can be scaled by a universal intensification factor.

The latest proposal binds each coil to a compatible irradiation module and introduces pure oxygen only at the Stage 2 inlet. For response Set 1, a 2 mL PFA coil in the Vapourtec UV-150 module is followed by a 20 mL FEP coil in the separately illuminated manual module. At a liquid feed of 0.020 mL min⁻¹, Stage 1 has a nominal liquid residence time of 100 min. The 0.100 M substrate feed corresponds to 0.00200 mmol min⁻¹; an oxygen inlet flow of 0.090 mL min⁻¹ at 273.15 K and 1 atm supplies 0.00402 mmol min⁻¹, or 2.01 equivalents. Both stages are proposed at 25 °C, with 450 and 448 nm irradiation, respectively. The Stage 2 inlet/STP index, $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$, is 181.82 min; it is not a measurement of residence under reactor pressure.

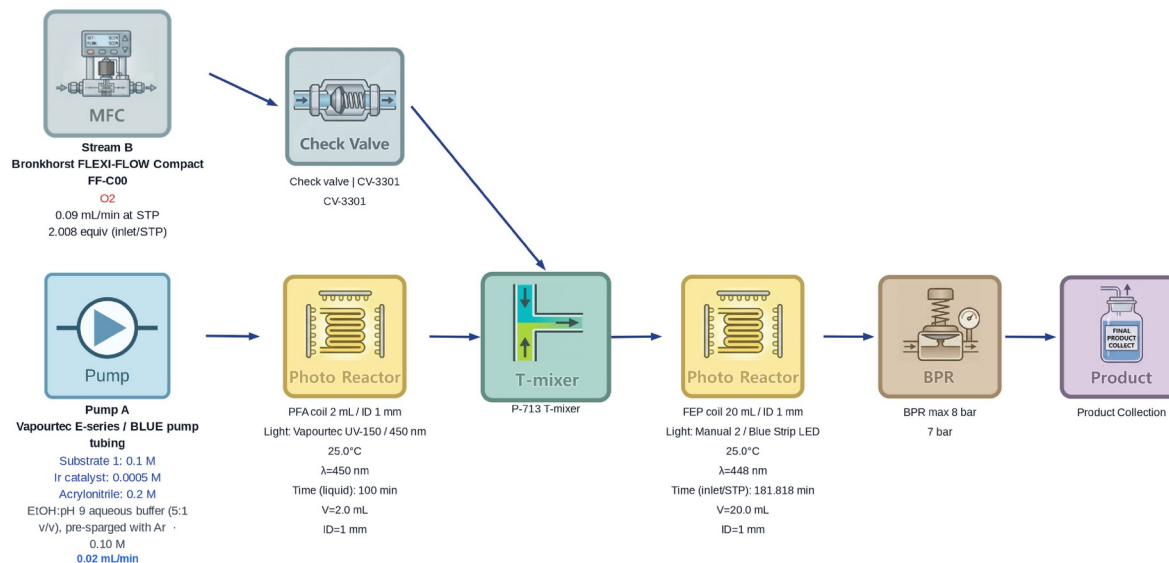
This revision responds to the collaborator's observation of gas entering the upstream reactor during an earlier high-gas-flow trial. Requiring pure oxygen lowers the total gas feed needed for a given oxygen molar input relative to air, but does not by itself resolve pressure-driven backflow. The archived proposal contains a gas-line check valve and a 7 bar(g) back-pressure setting; liquid-branch protection, pressure compatibility, and startup behavior still require laboratory review. The later council backflow tools provide conditional diagnostics, not a validated probability of failure or permission to execute. Figure S20 and Tables S29-S31 document the inputs, component feeds, three response-set outputs, and measurement plan. Sets 1 and 3 produce the same reported settings, whereas Set 2 increases the first-stage volume to 5 mL; these are not three independent replicates. Final sulfoxide yields are reserved as XXX, with conversion, selectivity, analytical method, and actual operating conditions to be entered alongside them.

a Protocol and design decisions

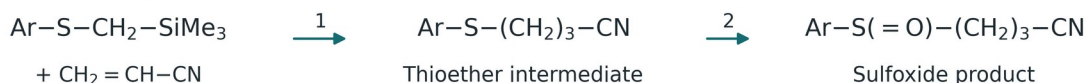
Batch: 4 h under Ar + 6 h in air, at 25 C.
Ar-sparged EtOH/pH 9 buffer (5:1); Ir, 0.5 mol%.
Substrate 0.10 M; acrylonitrile 0.20 M.

Oxygen-free feed to Stage 1; O₂ added at Stage 2.
Liquid 0.020; O₂ 0.090 mL/min (STP); 2.008 eq.
25 C; 450/448 nm; BPR 7 bar(g).

b Representative archived GUI topology | Set 1



c Chemical sequence



Abbreviated formulas: Ar = 4-methoxyphenyl; Me = methyl. Stage 2 adds pure O₂.

d Three response sets | archived conditions

Set	V1 (mL)	t1 (min)	V2 (mL)	t2, STP index (min)
1	2	100	20	181.82
2	5	250	20	181.82
3	2	100	20	181.82

t1 = V1/Qliquid; t2 = V2/(Qliquid + QO₂,STP), not measured contact time.

Sets 1 and 3 share conditions; they are not experimental replicates.

Measured yields pending: Set 1 XXX%; Set 2 XXX%; Set 3 XXX%.

Figure 5. Inventory-constrained photoredox Giese addition/oxidation. (a) Protocol and design context. (b) Archived Set 1 topology with compatible irradiation modules and oxygen introduced only before Stage 2. (c) Chemical sequence. (d) Proposed conditions for three response sets; XXX fields below reserve wet-lab yields, not measured or simulated data. Stage 1 time is V/Q_{liquid}; the Stage 2 value is an inlet/STP reporting index, not in-channel residence. Pressure/backflow checks remain unresolved. Full feeds, source inputs and other topologies are in ESI Section 8.1.

Case study 2: telescoped DPDTC-mediated amidation

The second case is a thermal, two-stage amidation of 3-methyl-4-nitrobenzoic acid with benzylamine (Figure 6). DPDTC and DMAP first generate a 2-pyridyl thioester; downstream aminolysis forms the amide C(O)-N bond while displacing the sulfur-containing leaving group.⁴⁶ The input specifies 30 min at 95 °C for each batch stage, separated by cooling and amine addition. In a telescoped flow proposal, the thioester is neither isolated nor assumed to form quantitatively. Its formation in Stage 1 limits the reactive intermediate available in Stage 2, making final amide yield the appropriate combined-process response.

The proposed feed structure retains the amine as a separate stream until the interstage junction. Feed A contains acid (0.500 M), DPDTC (0.525 M), and DMAP (0.0500 M) in 2-MeTHF; Feed B contains benzylamine (2.10 M). In Set 1, $Q_A = 0.160$ and $Q_B = 0.0400$ mL min⁻¹ give acid and amine molar flows of 0.0800 and 0.0840 mmol min⁻¹, preserving 1.05 equivalents of amine. The second reactor receives the sum of the two liquid feeds. Two 5 mL ETFE coils therefore give 31.25 and 25.00 min, respectively, at 95 °C. The 7 bar(g) setting is a proposed means of maintaining liquid operation above the solvent's normal boiling temperature; pressure ratings and thermal behavior must still be confirmed for the assembled train. Omitting batch cooling is a design hypothesis, not evidence that interstage temperature is irrelevant.

Sets 2 and 3 select the same reported operating point: a 10 mL first reactor and 5 mL second reactor with liquid feeds of 0.280 and 0.0700 mL min⁻¹, yielding stage times of 35.71 and 14.29 min. These proposals shorten the second stage while increasing the first-stage time, rather than applying a common speed-up factor to the complete sequence. The distinction can be tested by measuring residual acid, thioester, and final amide alongside the final yield. Feed concentrations are nominal preparation targets; premix stability, solubility, and the actual intermediate concentration require measurement. Figure S21 and Tables S32-S34 give the response sets, stage and feed calculations, and experimental reporting template. Final amide yields remain YYY pending insertion of the verified measurements and their analytical basis.

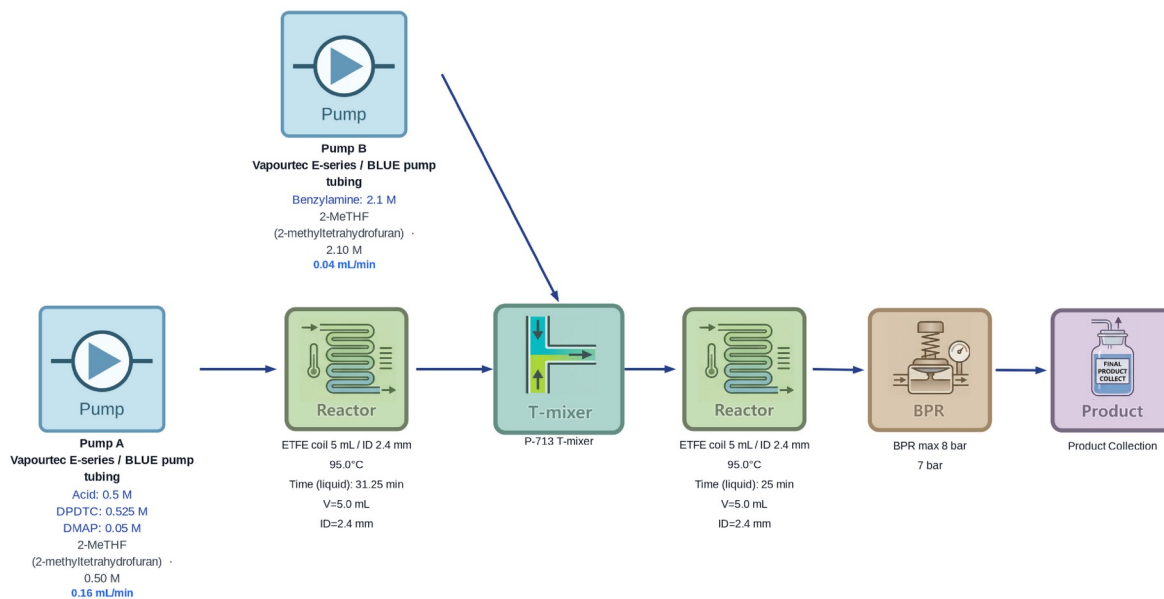
The alpha-bromination example from the original preprint⁴⁷ is retained unchanged as Figure S22 in ESI Section 9 for historical comparison. It is distinct from the current amidation case and is not presented as a new laboratory validation.

a Protocol and design decisions

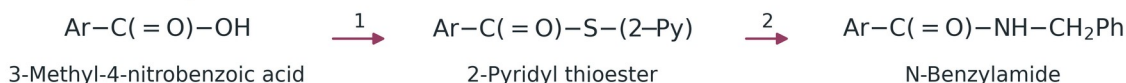
Batch: 30 + 30 min at 95 °C, with cooling between.
Feed A: acid 0.50 M; DPDTC 0.525 M; DMAP 0.05 M.
Benzylamine is introduced only at Stage 2.

Set 1: feed A 0.160 + feed B 0.040 mL/min.
Feed B: benzylamine 2.10 M (1.05 eq).
Two 5 mL ETFE coils; 95 °C; BPR 7 bar(g).

b Representative archived GUI topology | Set 1



c Chemical sequence



Abbreviations: Ar = 3-methyl-4-nitrophenyl; 2-Py = pyridin-2-yl; Ph = phenyl.

d Three response sets | archived conditions

Set	V1 / V2 (mL)	QA / QB (mL/min)	t1 / t2 (min)
1	5 / 5	0.160 / 0.040	31.25 / 25.00
2	10 / 5	0.280 / 0.070	35.7143 / 14.2857
3	10 / 5	0.280 / 0.070	35.7143 / 14.2857

t1 = V1/QA; t2 = V2/(QA + QB). All sets: 95 °C; BPR 7 bar(g).

Sets 2 and 3 share conditions; they are not experimental replicates.

Measured yields pending: Set 1 YYY%; Set 2 YYY%; Set 3 YYY%.

Figure 6. Connected two-stage DPDTC-mediated amidation. (a) Protocol and design context. (b) Archived Set 1 topology with downstream amine addition and no intermediate isolation. (c) Acid activation through a 2-pyridyl thioester to N-benzyl-3-methyl-4-nitrobenzamide. (d) Proposed stage-specific conditions; YYY fields below reserve wet-lab yields. Times follow V/Q for the specified liquid-phase design, not measured conversion; downstream flow includes both feeds. Sets 2 and 3 coincide and are not independent replicates. Feed concentrations, molar flows and reporting fields are in ESI Section 8.2.

DISCUSSION

The hybrid architecture separates chemical judgment from engineering arithmetic

The central contribution is a coordinated design workflow, rather than a claim that a larger number of model calls is intrinsically better. The literature and rule base supply context, deterministic calculations relate dependent quantities, and the council evaluates chemical and operational choices. The evaluations address these roles at different levels: Figure 3 measures retrieval alignment, Figure 4 compares delivered designs, and the internal ablations probe the contribution of review. Together, they support the use of a shared process record and explicit checks for batch-to-flow translation. They do not imply that a plausible chemical interpretation becomes experimentally correct merely because its arithmetic closes.

The archived failures explain why this distinction matters. Table S26 includes a one-shot residence-time value that conflicts with its stated basis and a FlowPilot preparation instruction that assigns a catalyst to both a feed and a packed bed. Table S22 records a disagreement between a final structured design and an accompanying generated narrative. These are different failure modes, so a favorable aggregate score cannot substitute for inspecting the underlying records. FlowPilot's practical value is to bring chemical assumptions, calculations, equipment assignments, and review decisions into an auditable workflow in which such conflicts can be identified and corrected. The same auditability is needed for evaluator mistakes, including the checked false positive in Table S26.

Positioning relative to prior AI systems for chemistry

Related AI systems address complementary parts of chemical research: ChemCrow³⁶ combines language-model reasoning with chemistry tools; Coscientist⁴⁰ investigates autonomous chemical research; El Agente³⁷ targets quantum-chemistry calculations; Zeng et al.³⁸ study multi-agent process optimization; and Materealize³⁹ addresses materials design and synthesis. Process simulation²⁴ and experimental optimization²⁶⁻³⁰ provide further capabilities for evaluating and refining operating conditions. FlowPilot focuses on the transition from a batch procedure and laboratory constraints to a first flow-process proposal. Its distinguishing emphasis is the joint handling of chemical interpretation, engineering consistency, equipment assignment, and traceable review. These systems are not evaluated head-to-head here, so this positioning describes task scope rather than a performance ranking.

The council contribution depends on model and configuration

The matched architecture comparison provides evidence of higher mean scores with FlowPilot for each of the five retained generators, with the size of the improvement depending on the model (Figure 4b). The

internal Qwen3.8-27B screen provides a complementary comparison: removing the council lowered the mean score from 0.928 to 0.670 across the three cases (Figure 4d; Table S10). Individual specialist removals produced smaller differences. Because each internal condition-case cell was generated only once, these results support descriptive component analysis rather than a statistically powered claim that every specialist is necessary or that the full configuration is always optimal.

The earlier model-matrix and candidate-budget experiments reveal additional configuration sensitivity (Figure S12). Some model combinations completed only a screening pathway rather than the validated pathway; all Gemma-upstream and GPT-4o-mini-upstream entries in Table S18 are marked screen required. Increasing the candidate budget broadened the explored design families and revision activity without producing monotonically better outcomes (Tables S19-S21). These observations qualify the role of the council: it can revise and compare proposals, but cannot be assumed to rescue every weak upstream result. They also distinguish historical development experiments from the matched model comparison in Figure 4.

Limitations and scope boundaries

Several limitations define the scope of the evidence. First, the three benchmark chemistries cover only part of flow-process design space, and three generation repeats provide a limited estimate of variability. The retained model cohort and archived software configurations also limit extrapolation to other models or releases. The matched comparison holds the task information constant, not the total inference budget, so it evaluates the delivered workflows rather than equal-compute performance. Figures 5 and 6 are reserved for testing selected proposals against laboratory observations; no reaction yield or experimental safety claim follows from the computational scores. Integration with experimental platforms⁴⁸ could extend that evaluation, but chemist review remains essential.

Second, gas-flow and residence-time definitions require careful interpretation. Gas equivalents are calculated from inlet molar flow at a declared reference temperature and pressure, not from a gas-to-liquid volumetric ratio. The nominal inlet/STP time, calculated from reactor volume divided by the combined liquid and inlet/STP gas flows, is also not a measured in-channel contact time. Gas compression, dissolution, phase holdup, and slip can separate these quantities.⁴⁹ The ESI retains the distinct time bases used by archived runs (Table S13); comparisons should use a common definition rather than silently relabeling old results.

Third, the engineering rule base (2,537 rules) is not uniformly dense across chemistry classes. Thermal synthesis and photoredox catalysis are well represented, but electrochemistry, biocatalysis, and gas-phase flow chemistry have sparser coverage, which may reduce audit-trail reliability in those domains.

Fourth, the outcome benchmark does not establish performance on long multistep telescopes. Figures 5 and 6 extend the evaluation toward connected two-stage chemistry, where an upstream conversion deficit changes the composition entering the next reactor. Independent stage yields, kinetic information, and operational observations may be needed to interpret final performance and guide subsequent refinement.

Fifth, LLM-based evaluation remains fallible even with fixed criteria and deterministic evidence. Shared model families, incomplete source information, or differences in response presentation may affect judgments. Identity masking is not complete architectural blinding, and multiple critical flags can concern the same underlying defect. The checked false-positive example in Table S26 illustrates why these judgments should remain auditable. Excluding a source paper from retrieval also does not establish that it was absent from model pretraining.

Outlook

The immediate next step is to use the existing feedback interface to connect proposed designs with measured process behavior, preserving the input, equipment profile, actual settings, and observed response at each iteration. This would support evidence-based refinement and future integration with Bayesian optimization tools²⁶⁻³⁰. Broader chemistry coverage and longer telescoped sequences are further directions for evaluation. Sustainability assessment would require measured or fully specified material and energy balances: reactor volume, nominal time, and temperature alone do not establish an E-factor or process mass intensity. These extensions build on the same auditable design record rather than assuming that a shorter or smaller process is necessarily better. Language-guided priors provide a related route for incorporating chemist hypotheses into experimental optimization while testing them against accumulating evidence⁵⁰; that separate method is a prospective integration opportunity, not an algorithm evaluated within the present FlowPilot benchmark.

CONCLUSION

FlowPilot connects a batch protocol and chemist-supplied constraints to a traceable first flow-process proposal. Standardized intake establishes the design context, chemistry-aware retrieval supplies precedents, deterministic calculation links dependent parameters, and specialist review assesses candidate

operating choices. Equipment reconciliation and a common final record connect the selected conditions to the process diagram and supporting report.

For the five-model, three-chemistry cohort, the mean fixed-criteria score increased from 0.794 for one-shot generation to 0.917 for FlowPilot, with fewer raw judge critical flags. Module ablations support a contribution from council review, while historical model-matrix and budget studies show that this contribution depends on configuration. These findings support the coordinated workflow as a practical basis for batch-to-flow design, within the limits of the evaluated cases and rubric. The Giese/oxidation and DPDTC-amidation studies will examine the separate question of experimental performance.

For the chemist, the intended benefit is a shorter path from an initial procedure to an inspectable proposal: enter the protocol, resolve the standardized questions, select an inventory profile, and review the linked design records. Measurements can then be returned with the actual conditions to inform a subsequent design cycle. This connects proposal generation with experimental development while preserving the distinction between what was assumed, calculated, proposed, and observed.

More broadly, the hybrid pattern demonstrated here, LLM-driven interpretation coupled with deterministic computation and multi-agent cross-domain audit, is not specific to flow chemistry. Any engineering design problem that is currently solved by a small team of complementary specialists iterating across coupled domains is a candidate for the same architectural approach. The modular design of FlowPilot is intended to make such extensions practical.

METHODS

A literature corpus containing 464 classified flow-chemistry records was annotated with available reaction, reactor, material, operating-condition, yield, and stream-count metadata. Field completeness was reported separately (Table S4). A complementary handbook-derived engineering knowledge base^{22-23,34} contained 2,537 rules categorized by topic and severity. The rule-base analysis describes coverage and recorded associations, while the deterministic calculator implements the numerical checks used during design.

The plan-aware retrieval pipeline first generates a chemistry plan from the user input and then constructs an expanded query containing reaction class, mechanism, photocatalyst family, and sensitivity flags. Retrieved candidates are reranked by a field-aware score that rewards mechanistic relevance over superficial semantic similarity.

The translation workflow comprises standardized intake, chemistry interpretation, retrieval, engineering calculation, candidate generation, council review, inventory realization, and topology rendering. LLMs propose chemical interpretations and independent design choices; deterministic routines recalculate dependent quantities and check their consistency. Chemist objectives and available evidence guide residence-time and throughput choices. A nominal intensification factor is not treated as proof of conversion, and an inventory-compatible operating point is not itself an experimentally validated procedure. Stage parameters, stream assignments, and topology are reconciled before final publication. Major development milestones and their dated evidence are summarized in Table S35 (ESI Section 10). Later scientific-policy and backflow extensions are separated from the frozen software used for the earlier architecture benchmarks.

The council reviews a candidate pool under bounded domain authority. Each of the seven agents (Designer, Chemistry Agent, Kinetics Agent, Fluidics Agent, Safety Agent, Skeptic, Orchestrator) may only modify fields within its own domain. Revised candidates are recomputed deterministically before final scoring.

The matched architecture comparison used the same frozen case input and final-output contract for one-shot generation and FlowPilot. Five retained generator models, two architectures, three chemistry cases, and three generation repeats yielded 90 outcomes in the ESI cohort. Three judge families received identity-masked final designs, source context, deterministic verification evidence, and the fixed rubric in Tables S8-S9. Each applicable criterion was rated from 0 to 4; ratings were averaged across judges and criteria, then divided by four. Gas bookkeeping and multistage closure were omitted only when inapplicable. Per-model error bars summarize the sample SD of three repeat-level means, each averaging the three cases. Critical flags were recorded separately. Exact inputs, model records, applicability rules, and archived evaluation limitations are given in ESI Section 3.

The internal module screen fixed the generator to Qwen3.8-27B and generated each condition-case combination once; its SD therefore measures variation between chemistries, not repeatability. Figure 4d shows selected conditions and Table S10 provides the displayed module summaries. The separate historical model-matrix experiment crossed four upstream and four council endpoints, and the candidate-budget study tested $B = 1, 6, 12$, and 24 with five repeats. Their validation dispositions, numerical data, and metric definitions are reported in ESI Section 6.1 and Tables S18-S21. These studies are kept distinct from the independent-judge outcome benchmark.

AUTHOR INFORMATION

Corresponding Author

*Gwang-Noh Ahn, E-mail: gnahn@kriect.re.kr

Author Contributions

A.M. and M.P. contributed equally to this work. The complete contributor-role statement and approval of the final author list will be confirmed by all authors before submission.

Notes

Availability of data and code

Source code and data will be provided to peer reviewers via GitHub during the review process and publicly released upon publication of this work.

ACKNOWLEDGMENT

This work was supported by the Strategic Research Program (“Accelerated Scale-up of Advanced Materials via Autonomous Experimental Systems”, DK2685-01) of the National Research Council of Science & Technology (NST), funded by the Ministry of Science and ICT (MSIT) of the Republic of Korea, and by the Research Program of the Korea Research Institute of Chemical Technology (KRICT) (KK2651-10 and BSK26-118).

ABBREVIATIONS

BPR, back-pressure regulator; DPDTC, dipyridyldithiocarbonate; Da, Damköhler number; FEP, fluorinated ethylene propylene; IF, intensification factor; LLM, large language model; MFC, mass flow controller; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STY, space-time yield.

REFERENCES

- (1) Rossetti, I. Flow Chemistry: New Concepts from Batch to Continuous Organic Chemistry. *Ind. Chem.* **2016**, 2, e102. <https://doi.org/10.4172/2469-9764.1000e102>.

- (2) Rossetti, I.; Compagnoni, M. Chemical Reaction Engineering, Process Design and Scale-up Issues at the Frontier of Synthesis: Flow Chemistry. *Chem. Eng. J.* **2016**, *296*, 56–70. <https://doi.org/10.1016/j.cej.2016.02.119>.
- (3) Feng, K.; Zhu, Y.-Y.; Gu, S.; Long, J.; Wang, H.-F. Typical Cases of Continuous Flow Chemistry in Pharmaceutical Synthesis in 2023-2024. *Asian J. Org. Chem.* **2025**, *14*, e202500132. <https://doi.org/10.1002/ajoc.202500132>.
- (4) Wegner, J.; Ceylan, S.; Kirschning, A. Flow Chemistry – A Key Enabling Technology for (Multistep) Organic Synthesis. *Adv. Synth. Catal.* **2012**, *354*, 17–57. <https://doi.org/10.1002/adsc.201100584>.
- (5) Cambié, D.; Bottecchia, C.; Straathof, N.; Hessel, V.; Noël, T. Applications of Continuous-Flow Photochemistry in Organic Synthesis, Material Science, and Water Treatment. *Chem. Rev.* **2016**, *116*(17), 10276–10341. <https://doi.org/10.1021/acs.chemrev.5b00707>.
- (6) Padoin, N.; Matiazzo, T.; Riella, H.; Soares, C. A Perspective on the Past, the Present, and the Future of Computational Fluid Dynamics (CFD) in Flow Chemistry. *J. Flow Chem.* **2024**, *14*, 239–256. <https://doi.org/10.1007/s41981-024-00313-4>.
- (7) Movsisyan, M.; Delbeke, E.; Berton, J.; Battilocchio, C.; Ley, S.; Stevens, C. Taming Hazardous Chemistry by Continuous Flow Technology. *Chem. Soc. Rev.* **2016**, *45* (18), 4892–4928. <https://doi.org/10.1039/c5cs00902b>.
- (8) Yasukawa, T.; Kobayashi, S. Translation of Batch to Continuous Flow in Photoredox Reactions. *ACS Central Science*. July 16, 2021, pp 1099–1101. <https://doi.org/10.1021/acscentsci.1c00711>.
- (9) Noël, T.; Cao, Y.; Laudadio, G. The Fundamentals Behind the Use of Flow Reactors in Electrochemistry. *Acc. Chem. Res.* **2019**, *52*, 2858–2869. <https://doi.org/10.1021/acs.accounts.9b00412>.
- (10) Gemoets, H.; Su, Y.; Shang, M.; Hessel, V.; Luque, R.; Noël, T. Liquid Phase Oxidation Chemistry in Continuous-Flow Microreactors. *Chem. Soc. Rev.* **2016**, *45* (1), 83–117. <https://doi.org/10.1039/c5cs00447k>.
- (11) Plouffe, P.; Macchi, A.; Roberge, D. From Batch to Continuous Chemical Synthesis—A Toolbox Approach. *Org. Process Res. Dev.* **2014**, *18*, 1286–1294. <https://doi.org/10.1021/op5001918>.

- (12) Ingham, R. J.; Battilocchio, C.; Fitzpatrick, D. E.; Sliwinski, E.; Hawkins, J. M.; Ley, S. V. A Systems Approach Towards an Intelligent and Self-Controlling Platform for Integrated Continuous Reaction Sequences. *Angew. Chem. Int. Ed.* **2015**, *54* (1), 144–148. <https://doi.org/10.1002/anie.201409356>.
- (13) Adamo, A.; Beingessner, R. L.; Behnam, M.; Chen, J.; Jamison, T. F.; Jensen, K. F.; Monbaliu, J.-C. M.; Myerson, A. S.; Revalor, E. M.; Snead, D. R.; Stelzer, T.; Weeranoppanant, N.; Wong, S. Y.; Zhang, P. On-Demand Continuous-Flow Production of Pharmaceuticals in a Compact, Reconfigurable System. *Science* **2016**, *352* (6281), 61–67. <https://doi.org/10.1126/science.aaf1337>.
- (14) Holtze, C.; Boehling, R. Batch or Flow Chemistry? – A Current Industrial Opinion on Process Selection. *Curr. Opin. Chem. Eng.* **2022**, *36*, 100798. <https://doi.org/10.1016/j.coche.2022.100798>.
- (15) Glasnov, T. N.; Kappe, C. O. The Microwave-to-Flow Paradigm: Translating High-Temperature Batch Microwave Chemistry to Scalable Continuous-Flow Processes. *Chem. Eur. J.* **2011**, *17* (43), 11956–11968. <https://doi.org/10.1002/chem.201102065>.
- (16) Chauhan, R.; Rana, A.; Mottafegh, A.; Kim, D. P.; Singh, A. K. Manual to Auto-Optimization Platform of Multistep Apixaban Synthesis. *Organic Process Research & Development* **2025**, *29* (3), 881–888. <https://doi.org/10.1021/acs.oprd.4c00535>.
- (17) Aand, D.; Rana, A.; Mottafegh, A.; Kim, D. P.; Singh, A. K. Autonomous closed-loop photochemical reaction optimization for the synthesis of various angiotensin II receptor blocker molecules. *Reaction Chemistry & Engineering* **2024**, *9* (9), 2427–2435. <https://doi.org/10.1039/D4RE00138A>.
- (18) Zhang, J.; Wang, K.; Teixeira, A.; Jensen, K.; Luo, G. Design and Scaling Up of Microchemical Systems: A Review. *Annu. Rev. Chem. Biomol. Eng.* **2017**, *8*, 285–305. <https://doi.org/10.1146/annurev-chembioeng-060816-101443>.
- (19) An, H.; Li, A.; Sasmito, A. P.; Kurnia, J. C.; Jangam, S. V.; Mujumdar, A. S. Computational Fluid Dynamics (CFD) Analysis of Micro-Reactor Performance: Effect of Various Configurations. *Chem. Eng. Sci.* **2012**, *75*, 85–95. <https://doi.org/10.1016/j.ces.2012.03.004>.
- (20) Ripken, R. M.; Wood, J. A.; Schlautmann, S.; Günther, A.; Gardeniers, H. J. G. E.; Le Gac, S. Towards Controlled Bubble Nucleation in Microreactors for Enhanced Mass Transport. *React. Chem. Eng.* **2021**, *6* (10), 1869–1877. <https://doi.org/10.1039/D1RE00092F>.

- (21) Mielke, E.; Plouffe, P.; Koushik, N.; Eyholzer, M.; Gottsponer, M.; Kockmann, N.; Macchi, A.; Roberge, D. M. Local and Overall Heat Transfer of Exothermic Reactions in Microreactor Systems. *React. Chem. Eng.* **2017**, *2* (5), 763–775. <https://doi.org/10.1039/C7RE00085E>.
- (22) Capaldo, L.; Wen, Z.; Noël, T. A Field Guide to Flow Chemistry for Synthetic Organic Chemists. *Chem. Sci.* **2023**, *14* (16), 4230–4247. <https://doi.org/10.1039/D3SC00992K>.
- (23) Plutschack, M. B.; Pieber, B.; Gilmore, K.; Seeberger, P. H. The Hitchhiker's Guide to Flow Chemistry. *Chem. Rev.* **2017**, *117* (18), 11796–11893. <https://doi.org/10.1021/acs.chemrev.7b00183>.
- (24) Haydary, J. *Chemical Process Design and Simulation*; Wiley: **2018**. <https://doi.org/10.1002/9781119311478>.
- (25) Reynoso-Donzelli, S.; Ricardez-Sandoval, L. A Reinforcement Learning Approach with Masked Agents for Chemical Process Flowsheet Design. *AIChE J.* **2025**, *71* (1), e18584. <https://doi.org/10.1002/aic.18584>.
- (26) Mottafegh, A.; Joo, J.-U.; Na, G.-S.; Sharma, V.; Kim, D.-P. AI-Assisted Autonomous Manufacturing of Tailored Drug-Loaded Nanoparticles by Multi-Step Continuous-Flow Platform. *Chem. Eng. J.* **2024**, *500*, 157454. <https://doi.org/10.1016/j.cej.2024.157454>.
- (27) Dunlap, J.; Ethier, J.; Putnam-Neeb, A.; Iyer, S.; Luo, S.-X. L.; Feng, H.; Torres, J. G.; Doyle, A.; Swager, T.; Vaia, R.; Mirau, P.; Crouse, C.; Baldwin, L. Continuous Flow Synthesis of Pyridinium Salts Accelerated by Multi-Objective Bayesian Optimization with Active Learning. *Chem. Sci.* **2023**, *14*, 8061–8069. <https://doi.org/10.1039/d3sc01303k>.
- (28) Ahn, G.-N.; Kang, J.-H.; Lee, H.-J.; Park, B. E.; Kwon, M.; Na, G.-S.; Kim, H.; Seo, D.-H.; Kim, D.-P. Exploring Ultrafast Flow Chemistry by Autonomous Self-Optimizing Platform. *Chem. Eng. J.* **2023**, *453*, 139707. <https://doi.org/10.1016/j.cej.2022.139707>.
- (29) Mottafegh, A.; Ahn, G.-N.; Kim, D.-P. Meta Optimization Based on Real-Time Benchmarking of Multiple Surrogate Models for Autonomous Flow Synthesis. *Lab. Chip* **2023**, *23* (6), 1613–1621. <https://doi.org/10.1039/D2LC00938B>.
- (30) Jorayev, P.; Soritz, S.; Sung, S.; Jeraal, M.; Russo, D.; Barthelme, A.; Toussaint, F.; Gaunt, M.; Lapkin, A. Machine Learning-Driven Optimization of Continuous-Flow Photoredox Amine Synthesis. *Org. Process Res. Dev.* **2025**, *29*, 1411–1422. <https://doi.org/10.1021/acs.oprd.4c00533>.

- (31) Tu, Z.; Choure, S.; Fong, M. H.; Roh, J.; Levin, I.; Yu, K.; Joung, J.; Morgan, N.; Li, S.-C.; Sun, X.; Lin, H.; Murnin, M.; Liles, J.; Struble, T.; Fortunato, M.; Liu, M.; Green, W.; Jensen, K.; Coley, C. ASKCOS: Open-Source, Data-Driven Synthesis Planning. *Acc. Chem. Res.* **2025**, *58* (11), 1764–1775. <https://doi.org/10.1021/acs.accounts.5c00155>.
- (32) Coley, C.; Green, W.; Jensen, K. Machine Learning in Computer-Aided Synthesis Planning. *Acc. Chem. Res.* **2018**, *51* (5), 1281–1289. <https://doi.org/10.1021/acs.accounts.8b00087>.
- (33) Hartog, P.; Westerlund, A.; Tetko, I.; Genheden, S. Investigations into the Efficiency of Computer-Aided Synthesis Planning. *J. Chem. Inf. Model.* **2025**, *65* (4), 1771–1781. <https://doi.org/10.1021/acs.jcim.4c01821>.
- (34) Guidi, M.; Seeberger, P. H.; Gilmore, K. How to Approach Flow Chemistry. *Chem. Soc. Rev.* **2020**, *49* (24), 8910–8932. <https://doi.org/10.1039/C9CS00832B>.
- (35) Ramos, M. C.; Collison, C. J.; White, A. D. A Review of Large Language Models and Autonomous Agents in Chemistry. arXiv November 14, 2024. <https://doi.org/10.48550/arXiv.2407.01603>.
- (36) M. Bran, A.; Cox, S.; Schilter, O.; Baldassari, C.; White, A. D.; Schwaller, P. Augmenting Large Language Models with Chemistry Tools. *Nat. Mach. Intell.* **2024**, *6* (5), 525–535. <https://doi.org/10.1038/s42256-024-00832-8>.
- (37) Zou, Y.; Cheng, A. H.; Aldossary, A.; Bai, J.; Leong, S. X.; Campos-Gonzalez-Angulo, J. A.; Choi, C.; Ser, C. T.; Tom, G.; Wang, A.; Zhang, Z.; Yakavets, I.; Hao, H.; Crebolder, C.; Bernales, V.; Aspuru-Guzik, A. El Agente: An Autonomous Agent for Quantum Chemistry. *Matter* **2025**, *8* (7), 102263. <https://doi.org/10.1016/j.matt.2025.102263>.
- (38) Zeng, T.; Badrinarayanan, S.; Ock, J.; Lai, C.-K.; Barati Farimani, A. LLM-Guided Chemical Process Optimization with a Multi-Agent Approach. *Mach. Learn. Sci. Technol.* **2025**, *6* (4), 045067. <https://doi.org/10.1088/2632-2153/ae2382>.
- (39) Kim, S.; Choi, J.; Jang, K.; Park, J.; Bernales, V.; Aspuru-Guzik, A.; Jung, Y. Materealize: A Multi-Agent Deliberation System for End-to-End Material Design and Synthesis. arXiv January 22, 2026. <https://doi.org/10.48550/arXiv.2601.15743>.
- (40) Boiko, D. A.; MacKnight, R.; Kline, B.; Gomes, G. Autonomous Chemical Research with Large Language Models. *Nature* **2023**, *624* (7992), 570–578. <https://doi.org/10.1038/s41586-023-06792-0>.

- (41) Chauhan, R.; Mottafegh, A.; Rana, A.; Singh, A. K. UV-visible Auto-Reactometry enabled parallel screening for accelerating chemical synthesis development with LLM. *Chemical Engineering Journal* **2026**, *533*, 174829. <https://doi.org/10.1016/j.cej.2026.174829>.
- (42) Guo, X.; Li, Y.; Kong, X.; Jiang, Y.; Zhao, X.; Gong, Z.; Zhang, Y.; Li, D.; Sang, T.; Zhu, B.; Jun, G.; Huang, Y.; Liu, Y.; Xue, Y.; Kundu, R. D.; Lim, Q. J.; Zhao, Y.; Granger, L. A.; Younis, M. B.; Keivan, D.; Sabharwal, N.; Sinha, S.; Agarwal, P.; Vandyck, K.; Mai, H.; Wang, Z.; Venkatesh, A.; Barik, A.; Yang, J.; Yue, C.; He, J.; Wang, L.; Xu, L.; Chen, H.; Wang, J.; Xu, L.; Shetty, R.; Guo, Z.; Song, D.; Jha, M.; Liang, W.; Yan, W.; Zhang, B.; Karnoor, S. B.; Zhang, J.; Pandya, R.; Gong, X.; Ganesh, M. B.; Shi, F.; Xu, R.; Zhang, Y.; Ouyang, Y.; Qin, L.; Rosenbaum, E.; Snyder, C.; Seiler, P.; Dullerud, G.; Zhang, X. S.; Cheng, Z.; Hanumolu, P. K.; Huang, J.; Kulkarni, M.; Namazifar, M.; Zhang, H.; Hu, B. Toward Engineering AGI: Benchmarking the Engineering Design Capabilities of LLMs. arXiv September 21, 2025. <https://doi.org/10.48550/arXiv.2509.16204>.
- (43) Norheim, J. J.; Rebentisch, E.; Xiao, D.; Draeger, L.; Kerbrat, A.; de Weck, O. L. Challenges in Applying Large Language Models to Requirements Engineering Tasks. *Des. Sci.* **2024**, *10*, e16. <https://doi.org/10.1017/dsj.2024.8>.
- (44) Ahn, G.-N.; Shin, J.-A.; Lee, S. Sustainable Continuous Synthesis of 3,5-Disubstituted Isoxazoles: Development of an Integrated Reaction–Separation–Recovery System for Deep Eutectic Solvents. *ACS Sustainable Chem. Eng.* **2025**, *13*, 6589–6597. <https://doi.org/10.1021/acssuschemeng.5c00580>.
- (45) Park, J.; Kim, S. H.; Cho, J.-Y.; Atriardi, S. R.; Kim, J.-Y.; Mardhiyah, H.; Park, B. Y.; Woo, S. K. Batch and Flow Synthesis of Sulfides and Sulfoxides Using Green Solvents and Oxidant through Visible-Light Photocatalysis. *Green Chem.* **2025**, *27*, 3284–3292. <https://doi.org/10.1039/d4gc05769d>.
- (46) Saunders, J. M.; Ocegüera Nava, E.; Li, J.; Wong, M. J.; Freiberg, K. M.; Lipshutz, B. H. Flow-to-Flow Technology: Amide Formation in the Absence of Traditional Coupling Reagents Using DPDTC. *ACS Sustainable Chem. Eng.* **2025**, *13*, 6646–6655. <https://doi.org/10.1021/acssuschemeng.5c00914>.
- (47) Lu, Y.; Yan, Z.; Jiang, Y.; He, H.; Zhou, J.; Gao, S.; Xing, D. α -Bromination of Aldehydes by Continuous Flow Chemistry and Its Application to the Flow Synthesis of 2-Aminothiazoles. *Organic Process Research & Development* **2025**, *29*, 2047–2055. <https://doi.org/10.1021/acs.oprd.5c00076>.

- (48) Coley, C. W.; Thomas III, D. A.; Lummiss, J. A. M.; Jaworski, J. N.; Breen, C. P.; Schultz, V.; Hart, T.; Fishman, J. S.; Rogers, L.; Gao, H.; Hicklin, R. W.; Plehiers, P. P.; Byington, J.; Piotti, J. S.; Green, W. H.; Hart, A. J.; Jamison, T. F.; Jensen, K. F. A Robotic Platform for Flow Synthesis of Organic Compounds Informed by AI Planning. *Science* **2019**, *365* (6453), eaax1566. <https://doi.org/10.1126/science.aax1566>.
- (49) Mallia, C. J.; Baxendale, I. R. The Use of Gases in Flow Synthesis. *Org. Process Res. Dev.* **2016**, *20* (2), 327–360. <https://doi.org/10.1021/acs.oprd.5b00222>.
- (50) Mottafeigh, A.; Ahn, G.-N. Adaptive human-in-the-loop optimization using language-guided priors for chemical experiments. *Journal of Chemical Information and Modeling* **2026**, *66* (15), 8833-8847. <https://doi.org/10.1021/acs.jcim.6c00976>.